

Image Captions for GitHub Repository

Project: Optical Alignment and Diffraction Pattern Analysis

Image 1: 0° Alignment

Diffraction pattern at 0° orientation, exhibiting high-intensity central laser spot and clearly defined rhombic satellite structures.

Image 2: 45° Orientation

Symmetry breaking observed at 45° rotation, showing pattern transition and redistribution of scattered light intensity.

Image 3: 90° Orthogonal View

Diffraction pattern at 90° orthogonal orientation, demonstrating directional filtering effects and minimal intensity distribution.

Image 4: Low Exposure (EV -1.5)

Reduced exposure setting (EV -1.5) to capture detailed satellite spot morphology without saturation of the central beam.

Image 5: Final Modulated Output

Composite visualization of the modulated laser output through the serpentine geometry, highlighting the interference of satellite modes.